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DATA SCIENCE × AI ENGINEERING
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DATA SCIENCE FIELD NOTES / 02B

DATA PREPARATION REPORT

Inspection, cleaning, feature engineering, encoding and feature screening

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Educational portfolio report. This work does not constitute a production lending model or lending policy.

Executive summary

This report documents the preparation of 614 historical loan-application records for later machine-learning study. The workflow inspects data quality, treats missing observations, removes the application identifier, creates two analytical features, encodes categorical variables, standardises selected numerical variables, assesses potential outliers and screens features for redundancy and exploratory relevance.

Verified result: the cleaned export contains 614 rows and 14 columns; the corrected numeric export contains 614 rows and 14 fully numeric columns. Both contain zero missing values and zero exact duplicate rows.

Dataset overview

Measure	Verified value
Applications	614
Original variables	13
Approved outcomes	422 (68.73%)
Not-approved outcomes	192 (31.27%)
Exact duplicate rows	0
Missing cells	149
Fields with missing data	7

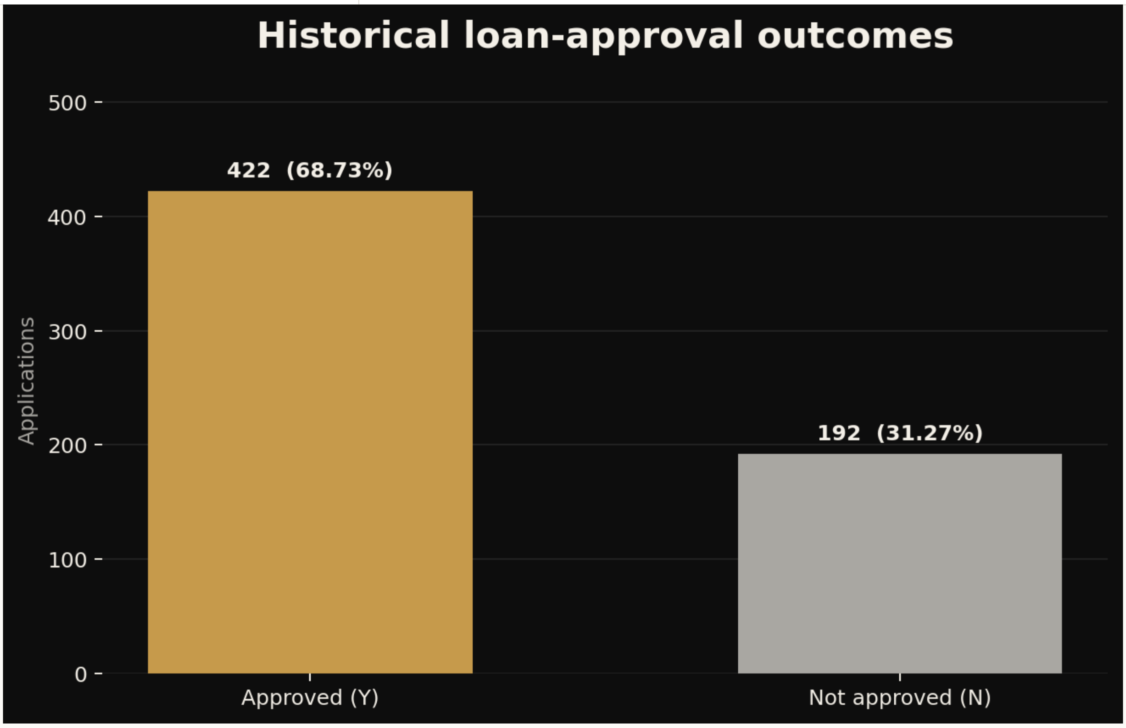

Figure 1. Historical outcome distribution in the source dataset.

Initial inspection

Inspection covered shape, column names, data types, summary statistics, target balance, missingness and exact duplicate rows. The application identifier is unique but does not provide an interpretable applicant characteristic, so it is removed before analytical preparation.

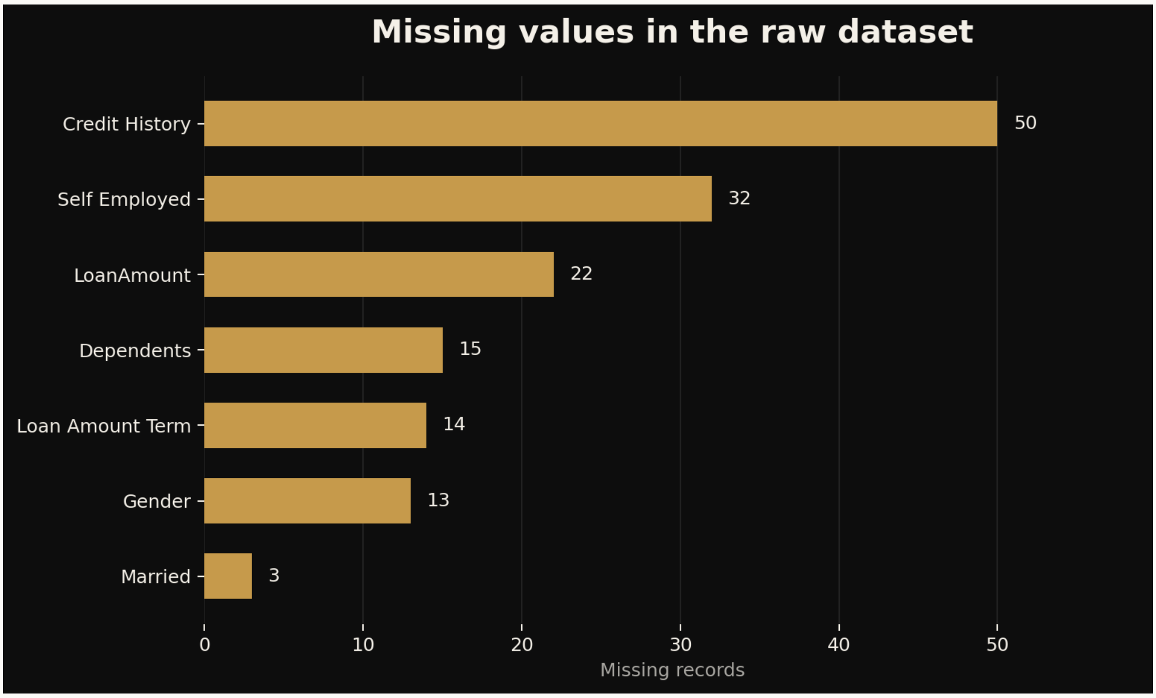

Figure 2. Missing-value counts before treatment.

Field	Missing	Treatment
Gender	13	Mode
Married	3	Mode
Dependents	15	Mode
Self_Employed	32	Mode
LoanAmount	22	Median
Loan_Amount_Term	14	Mode
Credit_History	50	Mode

Missing-value treatment

Low-frequency categorical missing values were filled using each field's most common category. LoanAmount was filled using the median because it is less sensitive to extreme values than the mean. Loan_Amount_Term and Credit_History were filled using their modes because both are discrete repeated fields in this dataset.

These are transparent educational choices, not universal defaults. In future model evaluation, every imputation rule must be learned from training data only and compared with suitable alternatives.

Feature engineering

Transformation	Definition	Reason
Remove Loan_ID	Drop the unique identifier	Avoid learning an arbitrary application code
total_income	applicant_income + coapplicant_income	Represent combined recorded household income
loan_income_ratio	loan_amount / total_income	Create a simple affordability-oriented indicator
snake_case names	Standardise all analytical column names	Improve readability and consistency in Python

No application had zero total income, so the ratio calculation did not create infinite values. The ratio remains a limited proxy because it excludes expenses, interest, debt and repayment capacity.

Categorical encoding

Field	Encoding
gender	Male = 1; Female = 0
married	Yes = 1; No = 0
education	Graduate = 1; Not Graduate = 0
self_employed	Yes = 1; No = 0
credit_history	Convert the 0/1 flag to integer
dependents	Convert 3+ to 3, then store as integer
property_area	One-hot encode with one reference category removed
loan_status	Y = 1; N = 0

Binary coding is used for compactness in this learning exercise. The numeric values do not imply that one social category is inherently better than another. Sensitive fields require necessity and fairness review before any real modelling proposal.

Numerical scaling

StandardScaler was applied to applicant income, co-applicant income, loan amount, loan term, total income and loan-income ratio. The corrected numeric export has means approximately equal to zero and population standard deviations equal to one for the retained scaled fields.

> The published numeric file records the full-sample preprocessing exercise. It must not be used to claim unbiased predictive performance because the scaler was fitted before a final evaluation split. Future work should place imputation, encoding and scaling inside a training-only pipeline.

Outlier assessment

Boxplots and the 1.5 x IQR rule were used to flag unusual values. The flagged observations were retained because the extreme income and loan values appeared plausible rather than clear data-entry errors. Removing valid extremes without domain evidence could erase important cases and distort the population represented by the practice data.

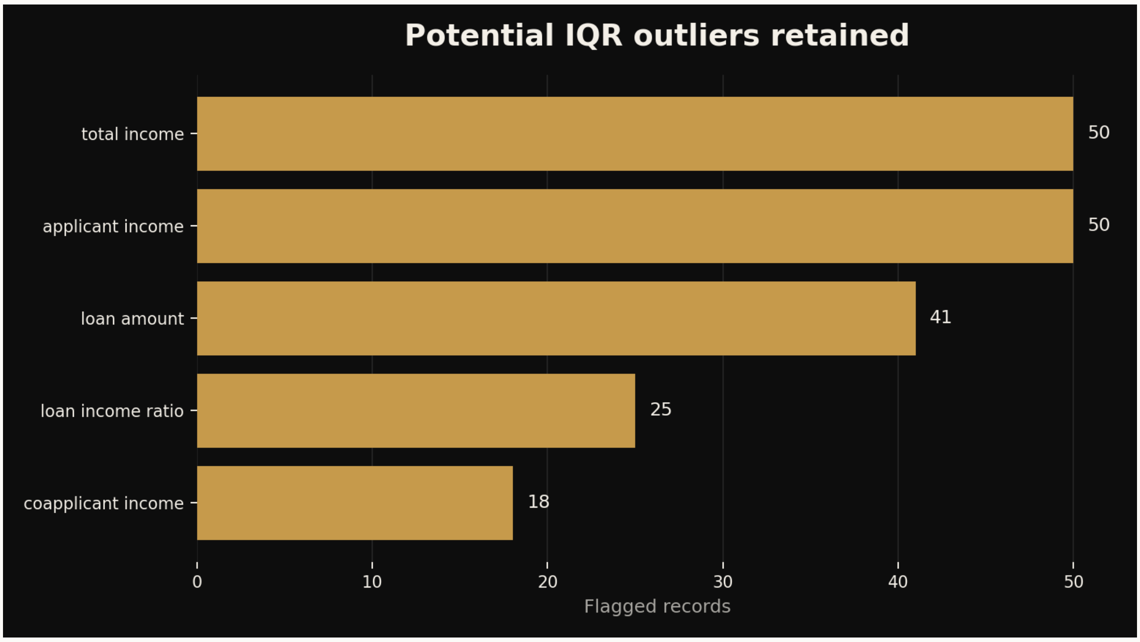

Figure 3. Potential outliers identified by the IQR rule and retained for analysis.

Feature screening

Feature screening combined correlation, target relationships, business logic and exploratory Random Forest importance. Applicant income and engineered total income showed a correlation of approximately 0.89. Applicant income was removed from the published numeric export to reduce redundancy while retaining the combined-income feature.

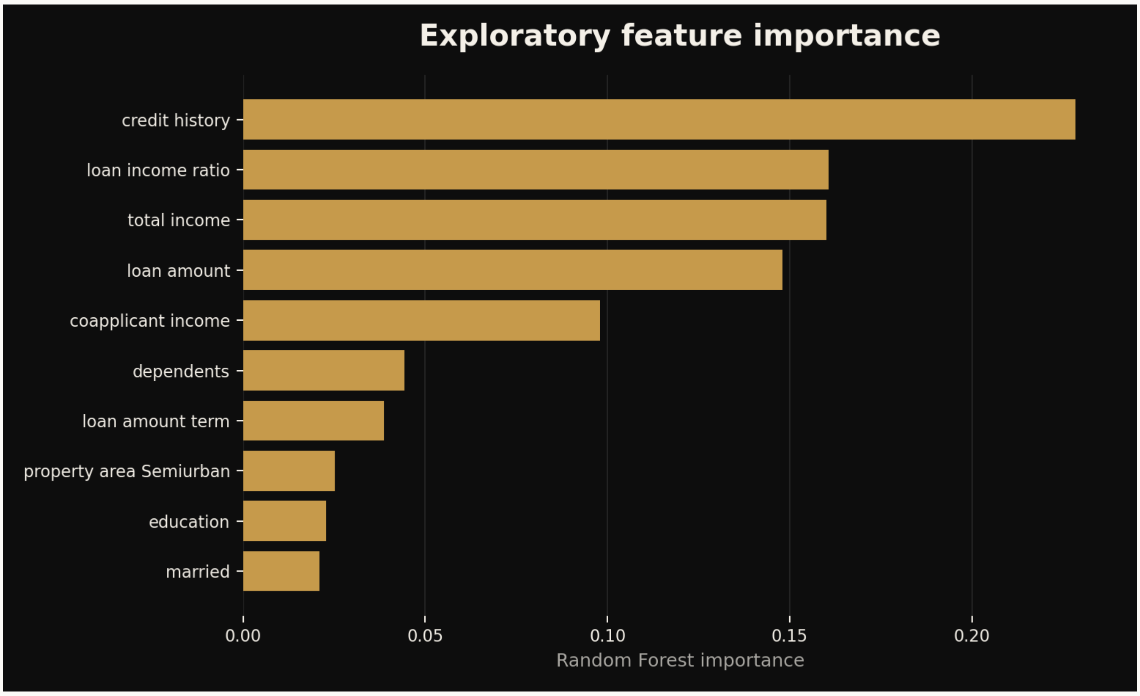

Figure 4. Exploratory Random Forest importance after the published feature reduction.

Credit history remains the strongest initial signal in the exploratory result. This importance is neither causal evidence nor proof that the field is sufficient, fair or suitable for real lending policy. Feature importance can change across splits, algorithms and preprocessing choices.

Published outputs

Output	Shape	Contract
loan_prediction_cleaned.csv	614 x 14	Human-readable, engineered, no missing values, no duplicate rows
loan_prediction_ml_ready.csv	614 x 14	Fully numeric, encoded, scaled full-sample demonstration artifact

The original notebook mistakenly exported the earlier pre-encoding dataframe under the ML-ready filename. The standardized repository corrects this by exporting the encoded and scaled dataframe and validating that every resulting column is numeric.

Reproducibility and verification

- The raw CSV is preserved unchanged in data/raw.
- scripts/prepare_data.py recreates both processed datasets.
- The notebook uses repository-relative paths and executes all code cells successfully.
- The preparation script validates shapes, missingness and numeric data types before publication.
- README charts and reports can be regenerated from versioned scripts.

Limitations and responsible use

- The practice dataset is small and does not establish representativeness.
- Mode imputation can reinforce majority categories and reduce variation.
- Historical approval labels may contain policy or social bias.
- Sensitive demographic variables require legal, ethical and fairness review.
- The full-sample numeric artifact is not suitable for unbiased performance evaluation.
- Feature importance and correlation describe association, not causation.
- No production model, deployment or real-world lending decision is claimed.

Conclusion

The data-preparation phase produced transparent, reproducible and internally consistent outputs suitable for continued educational modelling. The next phase should rebuild these operations in a scikit-learn Pipeline and ColumnTransformer, split before fitting any transformation, establish a transparent baseline, evaluate class-sensitive and calibration metrics, and examine subgroup errors before drawing practical conclusions.